THE STALK AT A GEOMETRIC POINT GIVES AN EQUIVALENCE FROM ABELIAN SHEAVES ON THE ETALE SITE OF A FIELD TO MODULES OVER THE ABSOLUTE GALOIS GROUP

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the equivalence for sets is quoted from the established theorem and the passage to abelian objects is justified by identifying both sides as categories of commutative group objects. There is no further step left to the reader.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Equivalence between sheaves of sets and Galois sets at a geometric point. Same chapter, Theorem theorem-equivalence-sheaves-point. Supplies the equivalence that is upgraded to abelian objects.
- Group objects in a category and transport along an equivalence. Standard: an abelian sheaf is a sheaf of sets equipped with a commutative group law, and a module over G is a G-set equipped with a commutative group law. Supplies the reason the equivalence restricts to abelian objects.

Source reference. Original theorem 03QT: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 8626–8626. Navigation pointer invoked at proof line 9092; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let $S = \mathrm{Spec}(K)$ with K a field. Let $\overline{s}$ be a geometric point of S. Let $G = Gal_{\kappa(s)}$ denote the absolute Galois group. The stalk functor induces an equivalence of categories*

$$Ab(S_{\acute{e}tale}) \longrightarrow Mod_G, \quad \mathcal{F} \longmapsto \mathcal{F}_{\overline{s}}.$$

Proof. In Theorem 03QT we have seen the equivalence between sheaves of sets and G-sets. The current lemma follows formally from this as an abelian sheaf is just a sheaf of sets endowed with a commutative group law, and a G-module is just a G-set endowed with a commutative group law. □